

Object Oriented Concepts || P2

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Java this Keyword

The this keyword in Java is a reference that refers to the current object of a class. It is available in instance methods, constructors, and instance initializer blocks, and it is commonly used to access the current object's instance members.

- Return the current object from a method.
- Pass the current object as an argument to a method or constructor.

Use of this Keyword:

- It is used to refer to the current class instance variable.
- It is used to refer to the current class method.
- It is used to refer to the current class constructor.
- It is used to pass the current class instance as a parameter to the method.
- It is used to pass the current class instance as a parameter to the constructor.
- It is used to return the current class instance from the method.

"super" Keyword in Java

- The super keyword in Java is used to refer to the immediate parent class object in an inheritance hierarchy. It allows a subclass to explicitly access parent class members when they are hidden or overridden. This keyword helps maintain clarity and control while working with inheritance.
- Used to call parent class constructors using `super()`.
- Helps access parent class methods and variables when overridden or hidden.
- Ensures proper inheritance behavior and code reusability.
The most common use of the super keyword is to eliminate the confusion between super classes and subclasses that have methods with the same name.

It can be used in two main ways:

- To access attributes and methods from the parent class
- To call the parent class constructor

"super" Keyword in Java

- **Use of super Keyword: It's a reference variable**
- It is used to refer the parent class instance variable.
- It is used to refer to the parent class method.
- It is used to refer to the parent class constructor.

Access Modifiers in Java

Access modifiers in Java are used to control the visibility and accessibility of classes, methods, and variables. They help enforce encapsulation by restricting access to different parts of a program. Java provides four types of access modifiers to define scope and protection levels.

- **Public modifier:** Accessible from anywhere in the program
- **Protected modifier:** Accessible within the same package and by subclasses
- **Private modifier:** Accessible only within the same class
- **Default modifier:** Accessible only within the same package

Access Modifiers in Java

1. Private Access Modifier

The private access modifier is specified using the keyword `private`. The methods or data members declared as private are accessible only within the class in which they are declared.

- Provides the highest level of restriction.
- Commonly used to protect fields and implementation details.

2. Default Access Modifier

If no access modifier is specified, this means only classes within the same package can access it.

- Provides more access than private but less than protected and public.
- Cannot be directly accessed from classes in other packages.

Access Modifiers in Java

- **3. Protected Access Modifier**

The protected access modifier is specified using the keyword `protected`. The methods or data members declared as protected are accessible within the same package or subclasses in different packages.

- Provides the least restrictive access.
- Commonly used for methods or classes intended to be accessed externally.

4. Public Access Modifier

- The public access modifier is specified using the keyword `public`. Public members are accessible from everywhere in the program. There is no restriction on the scope of public data members.

Access Modifiers in Java

Comparison Table of Access Modifiers in Java

	Default	Private	Protected	Public
Same Class	Yes	Yes	Yes	Yes
Same Package Subclass	Yes	No	Yes	Yes
Same Package Non-Subclass	Yes	No	Yes	Yes
Different Package Subclass	No	No	Yes	Yes
Different Package Non-Subclass	No	No	No	Yes